

Heart Failure 30-Day Readmission Prediction

Month 1 Project — Classification, evaluation, model comparison, and fit diagnosis

1. Project Objective

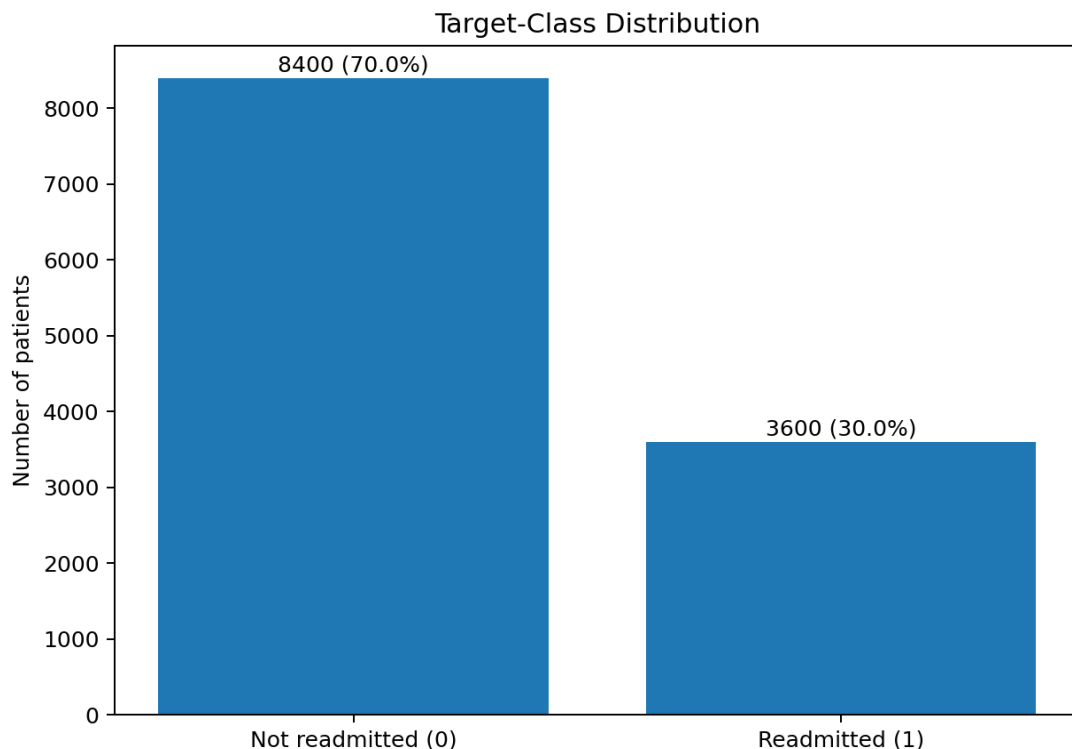
The supplied project brief requires three classical binary classification models—Logistic Regression, K-Nearest Neighbors (KNN), and Decision Tree—to predict 30-day readmission. The models must be compared using accuracy, precision, recall, F1-score and ROC-AUC, with confusion matrices and explicit train/test overfitting or underfitting analysis.

2. Dataset and Data Quality

The supplied dataset contains 12,000 patient records and 36 columns. The target is **Readmitted_30_Days**, encoded as 0 (not readmitted within 30 days) and 1 (readmitted within 30 days). The target distribution is 8,400 class-0 records (70%) and 3,600 class-1 records (30%), so the data are moderately imbalanced.

There are no missing values and no duplicate rows in the supplied dataset. Patient_ID is excluded because it is an identifier rather than a predictive clinical feature. Categorical variables are one-hot encoded. Numeric variables are median-imputed and standardized for Logistic Regression and KNN; the Decision Tree does not require scaling.

Leakage note: the model uses variables available in the supplied table, but real deployment requires a timestamp audit to verify that every predictor is known at the prediction point and does not contain information from after the prediction decision.



3. Experimental Design

The data were split 80/20 using stratification and random_state=42. All three models use the same underlying split. Preprocessing is implemented in scikit-learn pipelines so that imputation, scaling and encoding are fitted on training data only.

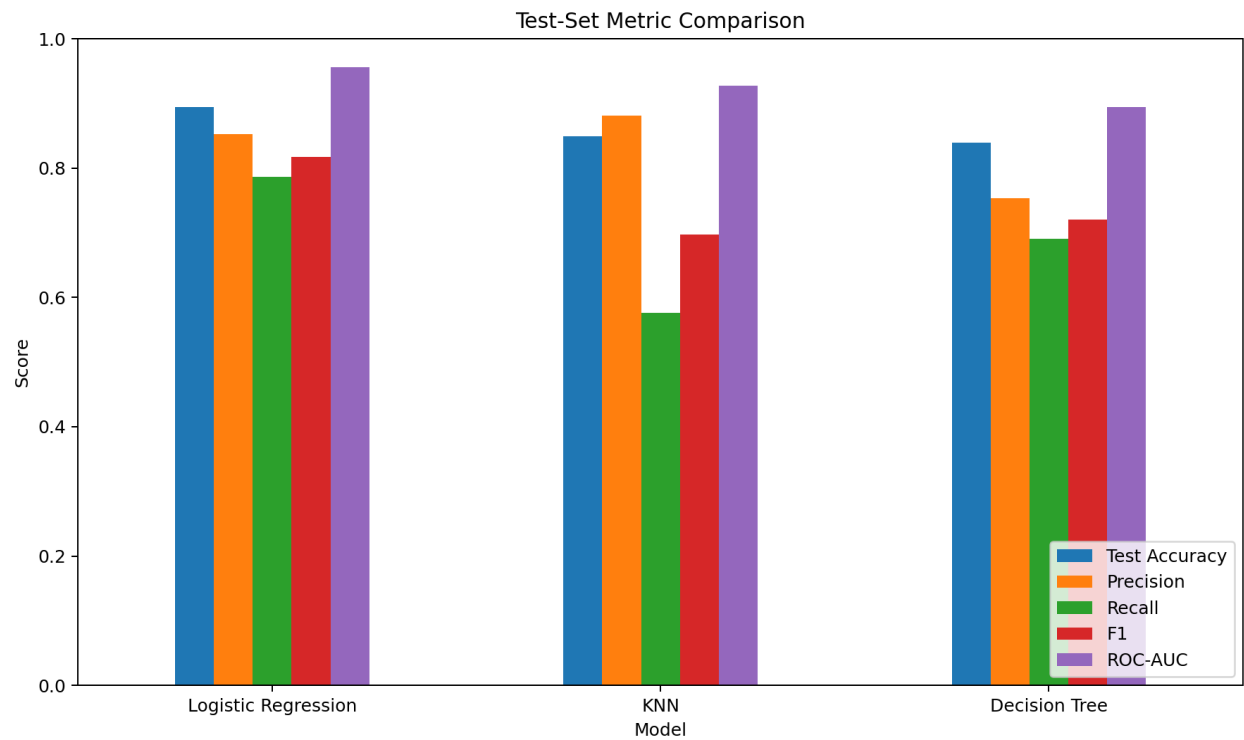
Hyperparameters were selected using 5-fold StratifiedKFold cross-validation on the training set with ROC-AUC as the selection metric. The final test set was not used for hyperparameter selection.

Selected settings

- Logistic Regression: C = 1.0
- KNN: k = 31
- Decision Tree: max_depth = 7, min_samples_leaf = 10

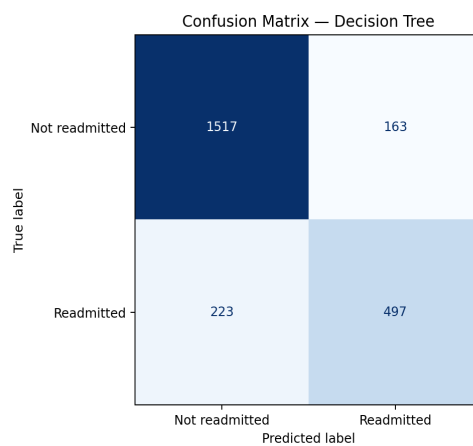
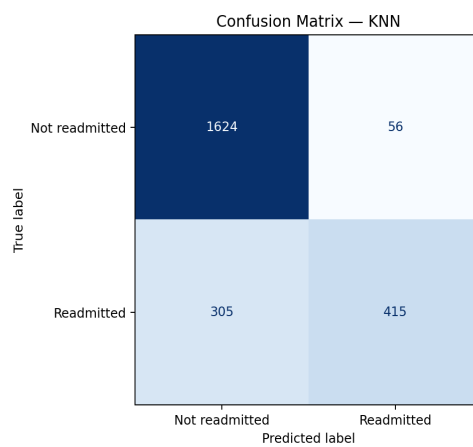
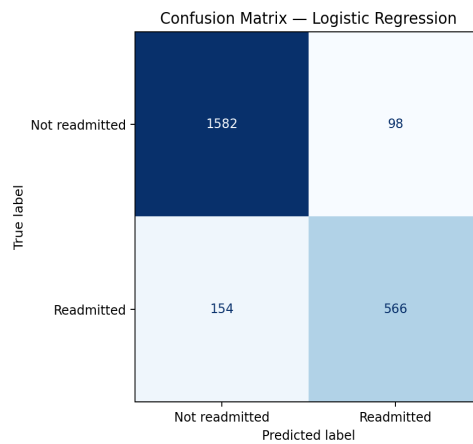
4. Final Model Comparison

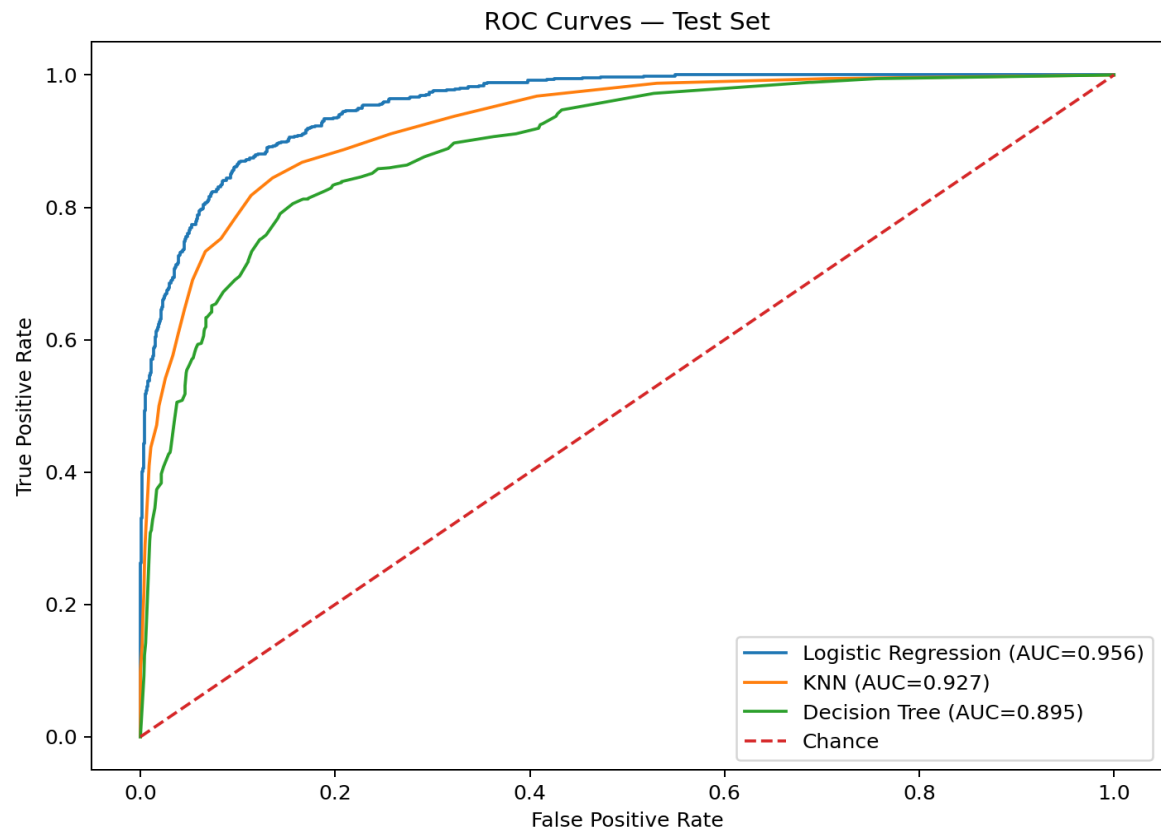
Model	Train Acc.	Test Acc.	Precision	Recall	F1	ROC-AUC	Fit diagnosis
Logistic Regression	0.895	0.895	0.852	0.786	0.818	0.956	Good generalization
KNN	0.855	0.850	0.881	0.576	0.697	0.927	Good generalization
Decision Tree	0.875	0.839	0.753	0.690	0.720	0.895	Mild generalization gap



5. Confusion Matrices and ROC Curves

For this task, a false negative is a patient who is actually readmitted within 30 days but is predicted as not readmitted. A false positive is a patient predicted as readmitted who is not readmitted. These error types have different operational consequences, so both precision and recall should be inspected.





6. Answers to Q1–Q14

Q1. What is the class distribution?

There are 8,400 class-0 records (70%) and 3,600 class-1 records (30%). The data are moderately imbalanced, so accuracy alone can conceal poor performance on the readmission class.

Q2. What preprocessing was performed?

Patient_ID was excluded. Categorical variables were one-hot encoded. Numeric variables were median-imputed and standardized for Logistic Regression and KNN. The Decision Tree used imputation and one-hot encoding without scaling. Pipelines ensure preprocessing is fitted on training data/folds only.

Q3. Why is KNN more sensitive to scaling?

KNN uses distances. A variable with a larger numeric scale can dominate those distances. Standardization places numeric features on comparable scales. Logistic Regression does not depend on distance, although scaling can still help optimization and regularization.

Q4. What k was used and how was it selected?

Five-fold cross-validation compared k values 3, 5, 9, 15, 21 and 31 using ROC-AUC. The selected value was $k=31$. Very small k values can have high variance, while very large k values can oversmooth local patterns.

Q5. What Decision Tree settings were used?

Cross-validation selected `max_depth=7` and `min_samples_leaf=10`. These constraints limit overly specific branches and improve generalization.

Q6. Compare the three models.

Logistic Regression has the strongest overall test profile: accuracy 0.895, precision 0.852, recall 0.786, F1 0.818 and ROC-AUC 0.956. KNN has higher precision but lower recall, while the Decision Tree has lower overall test ROC-AUC and F1.

Q7. Which metric deserves the most attention?

The answer depends on the cost of errors. If missing a true readmission is the main concern, recall is particularly important. If follow-up resources are scarce, precision also matters. A real deployment should select the probability threshold using clinical and operational costs.

Q8. Does any model appear overfit?

The Decision Tree shows the clearest generalization gap: training accuracy about 0.875 versus test accuracy about 0.839. This supports a mild overfitting/generalization concern. Logistic Regression and KNN have smaller gaps.

Q9. Does any model appear underfit?

No model shows strong evidence of classic underfitting because none has both very low training and very low test performance. KNN is weaker than Logistic Regression on several metrics, but that alone does not establish underfitting.

Q10. How could overfitting be reduced?

Reduce tree depth, increase minimum leaf/split constraints, and use cross-validation/pruning or feature selection. These measures reduce overly specific decision rules.

Q11. How could underfitting be improved?

If observed, reduce excessive regularization, increase model capacity, and engineer more informative features. For KNN, a smaller k increases local flexibility; for Logistic Regression, weaker regularization can allow more model flexibility.

Q12. Is highest accuracy automatically best?

No. Accuracy can be misleading with class imbalance and does not distinguish false-positive from false-negative costs. Precision, recall, F1 and ROC-AUC provide complementary evidence.

Q13. If ROC-AUC is similar but precision/recall differ?

Choose according to the operational cost of errors. Higher recall is appropriate when missed high-risk patients are especially costly; higher precision may be preferred when unnecessary interventions are costly. Threshold tuning can change this trade-off.

Q14. What are the limitations?

The dataset may not represent other hospitals or populations; feature timestamps must be audited for temporal leakage; and predictive performance does not establish clinical utility, calibration, fairness or safety. External validation and prospective evaluation are required before clinical deployment.

7. Final Conclusion

On the supplied dataset and the specified 80/20 stratified split, Logistic Regression provides the strongest overall test performance and a very small train/test accuracy gap. The Decision Tree shows the clearest generalization gap, while KNN trades higher precision for lower recall. This is a model-development result for the supplied dataset and should not be interpreted as evidence of clinical readiness.

Required submission components are provided with this report: a reproducible Jupyter notebook, metric CSV, plots, and README.